Rong Wang
Duke University
Fuqua School of Business
Email: rong.wang1996@duke.edu
Phone: +1 (919) 597-8388

Research Statement

My primary research fields are empirical asset pricing and investment, focusing on risk pricing, financial market frictions, and the design of investment strategies. I also study how fiscal policy interacts with macroeconomic conditions and financial markets, analyzing the mechanisms through which fiscal shocks drive inflation and affect asset prices. Methodologically, my work employs advanced empirical methods and large-scale data analysis to address interesting questions in asset pricing and macro-finance.

Empirical Asset Pricing and Investment

My current research investigates zero-beta rates and zero-beta (factor-neutral) investment strategies. In my job market paper, **"Resolving the Zero-Beta Rate Puzzle"**, I show that the persistently high estimated zero-beta rates across asset pricing models arise from pervasive factor model misspecification rather than a genuinely high risk-free rate or convenience yield. When a factor model does not perfectly price assets, the implied zero-beta rate becomes non-unique and tends to be upward biased. I introduce a new investment-based measure of misspecification—the maximum Sharpe ratio attainable by zero-investment, zero-beta portfolios—which quantifies the economic magnitude of pricing errors. Using both classical and machine-learning factor models, I find substantial and systematic misspecification, explaining their uniformly elevated zero-beta rates. Simulations confirm that realistic levels of misspecification can fully reproduce the empirical puzzle even when the true risk-free rate is low. Together, these results reinterpret the zero-beta rate puzzle as a statistical artifact of imperfect factor pricing, offering new insights on the risk-free rate, risk premium, and convenience yield.

In a related paper, **"Factor Investing and Factor-Neutral Investing"**, examine the practical performance of machine-learning (ML)–based factor investing strategies. While such methods promise high out-of-sample Sharpe ratios, they consistently fail to deliver in practice. I show that factor mean–variance portfolios constructed from leading ML models are economically unviable, generating strongly negative net returns once realistic trading costs are incorporated. In contrast, factor-neutral investing—which constructs optimal zero-beta portfolios to exploit model mispricing rather than harvest factor premia—proves far more robust and implementable. The zero-beta constraints act as a natural shrinkage device, reducing turnover and enhancing resilience to trading frictions. In a comprehensive out-of-sample backtest covering 1990–2024, factor-neutral strategies deliver net-of-cost annualized Sharpe ratios between 0.6 and 1.2 and monthly alphas of 0.4%–1.5%, unexplained by known factors. These results imply that, in practice, the most valuable signal in ML-based factor models is not their estimated factors, but their errors.

Building on my previous findings that even state-of-the-art factor models—including those based on advanced machine-learning techniques—remain substantially misspecified, I explore the sources of these misspecifications in a new project, **"What Has Gone Wrong in the Best Factor Models"**. In this paper, I use zero-beta portfolios as diagnostic tools to trace where risk pricing breaks down. By analyzing the behavior of zero-beta returns across model specifications, asset groups, and market conditions, I aim to identify whether the underlying issues stem from missing risk factors, model instability, non-linear pricing, or limits to arbitrage. The goal is to move beyond documenting the existence of misspecification toward understanding its economic origins, offering new insights into why modern factor models, despite their sophistication, continue to leave large and systematic pricing errors unexplained.

Fiscal Policy and Fiscal Shocks

A second branch of my research studies the interaction between fiscal policy, macroeconomics, and financial markets. In **"Fiscal Policy and the Government Debt Maturity Structure"**, I use vector autoregressions (VAR) with external instruments to identify fiscal shocks and examine how their macroeconomic effects depend on the government debt maturity structure using local projections. I find that longer debt maturity significantly attenuates the output and inflation responses to fiscal expansions. Using a structural Fiscal Theory of the Price Level (FTPL) model, I show that longer maturity enables the government to realize capital gains on outstanding long-term debt after a deficit shock, reducing the incentive to inflate away existing liabilities. This mechanism explains the state-dependent transmission of fiscal policy observed in the data.

In a new and ongoing project, **"Fiscal Shocks, Beliefs, and Asset Prices"** (with Anna Cieslak and Hao Pang), we develop new narrative, high-frequency fiscal policy shocks that cover a wide range of fiscal actions, including government budgeting, legislation bills, sovereign credit rating events, debt limit discussions, Treasury auctions, and defense spending, among others. Using these shocks, together with textual analysis of a large corpus of news articles, we investigate the impact of fiscal policy on financial markets.